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ORIGINAL RESEARCH

Insights into the critical role of anions in nanofibrillation of cellulose in deep eutectic solvents

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Abstract Complex interactions between cellulose molecules and small molecules of deep eutectic solvents (DESs) can lead to dramatic changes in the structure of the hydrogen bond network in cellulose. However, the molecular and ionic evolution mechanism of DESs on cellulose nanofibrillation is still unclear. In this study, the mechanism and the critical role of ions in DESs on cellulose nanofibrillation were investigated. The results showed that cellulose nanofibers (CNFs) with a diameter of less than 200 nm could be prepared directly using DES consisting of lactic acid (LA) and chloride, e.g., LA/ammonium chloride (NH_4Cl) or LA/dodecyltrimethylammonium chloride (DTAC). Using LA/Zinc chloride ($ZnCl_2$), CNFs with a diameter of less than 100 nm can be prepared. The Kamlet-Abboud-Taft parameters (KAT value) showed that DESs containing LA had higher hydrogen bond acidity (α) than other systems (such as ethylene glycol and urea) without LA, which provided more active protons to improve the proton mobility in the cellulose nanofibrillation. Also, DESs composed of LA and Cl^- had high hydrogen bond alkalinity (β), which led to great changes in the original hydrogen bond network of cellulose and the formation of a new hydrogen bond structure, thereby easing the dissolution of the cellulose. Thus, the cellulose hydroxy groups could be esterified by the LA under high-temperature moistening conditions, and the CNFs prepared by the DES system containing LA and Cl^- had the highest degree of substitution (DS) (up to 0.73 in LA/$ZnCl_2$ system) after high-temperature moistening.

Keywords Cellulose nanofibrillation · Deep eutectic solvent · Anions · Cations · KAT value

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Introduction

Cellulose is a renewable resource with abundant energy storage on Earth. Cellulose is found in microfibril in wood and plant cell walls and serves as a structural scaffold for cells in green plants. Deconstructing fibers from wood or other structures formed by plants can produce cellulose nanofibers (CNFs) and/or cellulose nanocrystals (CNCs). They have the characteristics of high specific surface area, high aspect ratio, adjustable pore structure, and excellent

biocompatibility and biodegradability. It is widely used in biomedicine, nanocomposite materials, environmental protection, and optoelectronic devices (Zhu et al. 2016; Kontturi et al. 2018; Du et al. 2019; Guan et al. 2021; Chen et al. 2023b). During biosynthesis, glucan chains coalesce into extended structures—the elementary fibrils. These fibrils are often perceived as being constituted by a crystalline core covered by disordered (amorphous) surface chains. As a consequence of having its hydroxyl groups equatorially positioned a cellulose polymer can form hydrogen bonds between sequential glucose units: between the hydroxyl group on C3 and the ring oxygen (O3H3···O5) and between the hydroxyl groups on C2 and C6 respectively (O2H2···O6 or O6H6···O2) (Wohlert et al. 2021). Due to the rich hydroxyl group in cellulose, the close arrangement of hydrogen bonds between and within the chains of cellulose molecules is promoted, which also makes the deconstruction of cellulose fibers complicated. The key method to achieve efficient preparation of nanocellulose is to preferentially weaken the hydrogen bond network of cellulose to simplify the deconstruction of cellulose fibers.

Deep eutectic solvents (DESs) are garnering significant attention in cellulose nanofibrillation and biomass pretreatment, which are composed of hydrogen bond acceptor (HBA) and hydrogen bond donor (HBD) in a specific stoichiometric ratio, and have several advantages such as low cost of preparation, low toxicity, and excellent biodegradability (Abbott et al. 2003; Procentese et al. 2015; Smith et al. 2014). Notably, some DESs possess the remarkable ability to functionalize the surface of cellulose while effectively moistening the cellulose, which makes it efficient to modify cellulose and enhance the application to various applications (Tang et al. 2017). DESs exhibit the ability to form a robust internal network of hydrogen bonds, thereby efficiently moistening the cellulose fibers and disrupting the existing hydrogen bonding within the cellulose structure (Shen et al. 2019; Yang et al. 2019; Hong et al. 2020). Furthermore, research has proved that treating cellulose with acidic DESs can cause the hydrolysis of the amorphous regions and introduce carboxyl groups onto the cellulose surface, resulting in the formation of corresponding ester bonds and generating electrostatic repulsion, thus, the hydrophobic properties of cellulose are enhanced. It was found the carboxylated nanocellulose showed excellent dispersibility in aqueous solution and good thermal stability, which was conducive to its application in the field of composite materials (Ji et al. 2019; Liu et al. 2021).

However, there are few reports on the interaction mechanism between the cellulose and components in DESs, and the mechanism of cellulose nanofibrillation in DES systems is still unclear. Therefore, determining which component of DESs is more critical for cellulose nanofibrillation is an important scientific question (Mota-Morales et al. 2018; Sharma et al. 2023). Hydrogen bonds play a crucial role in cellulose, impacting its structure, stability, and properties. However, cellulose nanofibrillation involves more complex interactions, besides hydrogen bonds, other forces such as van der Waals forces, hydrophobic interactions, and electrostatic forces also affect the nanofrillation of cellulose. These interactions influence the dispersion stability and anti-aggregation properties of cellulose nanofibers (Wohlert et al. 2021). The dissolution mechanism of cellulose in ionic liquids (ILs) has been discussed (Wang et al. 2012). Most studies show that there is a linear relationship between the solubility of cellulose in ILs and the hydrogen bond alkalinity of anions, while the interaction between cations and cellulose is not strong (Li et al. 2018). However, there are still many controversies (Lu et al. 2014; Endo et al. 2016). The nanofibrillation of cellulose using DESs and the dissolution process of cellulose are somehow similar, both involving the interaction between cellulose and solvent and the destruction of hydrogen bond networks (Dias et al. 2020; Bi et al. 2023). Many studies have reported that almost all DESs are nontoxic, stable, inexpensive, environmentally friendly, and easy to synthesize (Chen and Mu 2019), which has a wider application prospect than ILs. Therefore, it is of great significance to explore the mechanism of cellulose nanofibrillation in DESs.

In the previous study, we also verified that carboxylic acid-based DESs could achieve cellulose nanofibrillation very efficiently and obtain high-yield CNFs, and the lactic acid/choline chloride DES could be used to achieve the preparation of CNFs and LCNFs (with a high lignin content) (Liu et al. 2021; Zhang et al. 2021). Based on this, different DESs with different HBDs and different anions or cations were selectively formulated to determine the role of each component in DESs, and to provide theoretical

guidance for the selection of nanocelluloses preparation system.

Materials and methods

Materials and reagents

Hardwood bleached kraft pulp as the cellulose sample was provided by Jiangsu Oji Paper Co., Ltd. Lactic acid (LA, CAS no.50-21-5, AR, 85-90%), ethylene glycol (EG, CAS no.107-21-1, AR, 98%), urea (Ur, CAS no.57-13-6, AR, 99%) were purchased from Sinopharm Group Chemical Reagent Co., Ltd. Choline hydroxide (ChOH, CAS no.123-41-1, AR, 44wt% in H_2O), choline dihydrogen citrate (Ch-Ca, CAS no.77-91-8, AR, 98%), choline chloride (ChCl, CAS no.67-48-1, AR, 98%), ammonium chloride (NH_4Cl, CAS no.12125-02-9, AR, 99.5%), dodecyltrimethylammonium chloride (DTAC, CAS no.112-00-5, AR, 99%), choline bromide (ChBr, CAS no.1927-06-6, AR, 98%) were purchased from Shanghai Aladdin Chemicals Company, zinc chloride ($ZnCl_2$, CAS no.7646-85-7, AR, 98%) was purchased from Nanjing Chemical Reagent Co., Ltd., and copper ethylenediamine solution was purchased from China National Pulp and Paper Research Institute. All reagents have not been purified.

Formulation of different DESs consisted of different HBD and HBA (composed of different anions or cations)

Three kinds of DESs were prepared to identify the key factors affecting the cellulose nanofibrillation. The pure LA and H_2O/ChCl system was treated as a control, and the DESs composed of different groups of HBD and ChCl (such as LA/ChCl, Ur/ChCl, EG/ChCl) were used to determine the role of organic carboxylic acid in the cellulose nanofibrillation. Analogously, DESs composed of different cations of HBA and LA (LA/NH_4Cl, LA/$ZnCl_2$, LA/DTAC) and DESs composed of different anions of HBA and LA (LA/ChBr, LA/Ch-La, LA/Ch-Ca), to determine the role of chloride and choline cations in cellulose nanofibrillation. All of the DESs are controlled by strict proportions and are simply mixed to form uniform and stable DESs for subsequent cellulose nanofibrillation. The preparation of LA/Ch-La requires LA to react with ChOH in advance, then remove the water in the system, add additional LA, and finally control the molar ratio of LA/Ch-La to 9/1. Detailed data of DESs composed of different HBDs and HBAs of different anions and cations are shown in Table 1. The viscosities of different DESs were detected using hydrometers and a Haake Rheostress 6000 (Thermo, Germany) at room temperature. The viscosity of different DESs is presented in Table 1.

Preparation of CNFs or cellulose microfibers (CMFs)

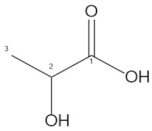
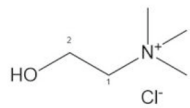
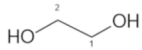
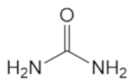
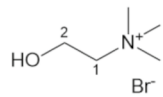
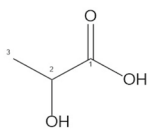
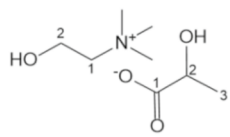
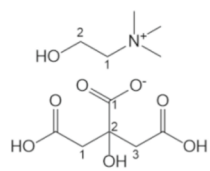
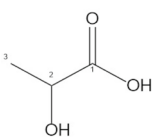
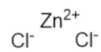
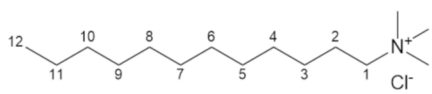
The cellulose samples were broken into small pieces using a fiber dissociator (GX-03, Gaoxin and Industry & Trade Co., LTD., China) swelled with different DES (mass ratio of cellulose: DESs is 1: 10) at 100 ℃ with continuous stirring for 3 h. After swelling, the neutral mixtures were then mechanically treated by deep grinding using a colloid mill (JMF-65, Shanghai Sobe Fluid Machinery Co., Ltd., China) at a rotor speed of 3000 rpm for 40 min (Zhang et al. 2021). Each cellulose sample was treated with different DES swelling, and other experimental conditions were kept the same. Finally, CNFs, or cellulose fibers prepared by different DESs were obtained.

Analyses and characterizations of CNFs or CMFs

Micromorphology characterization

Scanning electron microscopy (SEM) was used to observe the morphologies of the samples. The CNFs or CMFs were dialyzed with deionized water to remove residual reagent impurities, then diluted, and dropped onto a silicon single-crystal polished sheet, the water was removed by vacuum drying. Gold was sprayed onto the silicon wafer at a spraying current of 20 mA and a spraying time of 90 s. The morphologies of the samples were observed using a field emission electron microscope model JEOL-JSM7600F. The morphologies of the obtained CNFs were observed using transmission electron microscopy (TEM, JEOL EM-1400) at an acceleration voltage of 80 kV. Before imaging, a drop of 0.01% (w/w) CNF suspension was deposited onto an electron microscope glow-discharged

Table 1 DESs composed of different HBD and HBA of different anions and cations

DESs	HBD	HBA	Molar Ratio	Viscosity (Pa. s)
LA/ChCl			9 : 1	0.0658
EG/ChCl			1 : 1	0.0150
Ur/ChCl			2 : 1	149.6327
LA/ChBr			9 : 1	0.1043
LA/Ch-La			9 : 1	0.2861
LA/Ch-Ca			9 : 1	0.2902
LA/NH_4Cl			9 : 1	0.1286
LA/$ZnCl_2$			9 : 1	0.1147
LA/DTAC			9 : 1	0.0830

copper grid with a carbon-reinforced Formvar film and maintained on the grid for 2 min and then dry.

Particle size analysis

The particle size distributions of all the CNF samples were analyzed using a BeNano 90 laser diffraction particle size analyzer (Dandong Berrersize Ltd., China) with automatic analysis software. The CNF suspensions (0.2% w/v), previously sonificated for 30 s, were prepared and analyzed to determine the length.

Fourier transform infrared spectroscopy (FT-IR)

The samples were dried in a vacuum drying oven before testing. The FT-IR spectra of the sample were measured on the spectrometer (Thermo Scientific Nicolet iS5 with iD7 ATR accessory, USA) within the wavenumber region of 400 to 4000 cm^{-1}.

X-ray Photoelectron Spectroscopy (XPS)

The chemical state of element C of cellulose and CNFs was determined with XPS analysis. The XPS spectra were recorded on an ESCA instrument (Thermo Scientific K-Alpha). The approximate degree of surface substitution (DS) on each glucose unit of CNFs was calculated with the corresponding O−C=O group concentration, which was expressed by the following formula (Zhou et al. 2018):

$$DS = \frac{\mathbf{C}(\mathrm{O}-\mathrm{C}=\mathrm{O})/2}{\mathbf{C}(\mathrm{C}-\mathrm{O})/5}$$

where **C**(C−O) and **C**(O−C=O) represent the concentrations of C−O and O−C=O groups.

X-ray diffraction analysis (XRD)

The crystal structure of cellulose and CNFs was measured on a Bruker D8 ADVANCE X-ray diffraction (XRD) spectrometer under 40 kV and 30 mA. The diffraction angle 2θ was from 5 to 40° with a testing speed of 5°/min. According to Segal eq., the crystallinity index (CrI) of the samples can be calculated as follows:

$$\mathrm{CrI}(\%) = \frac{I_{200} - I_{am}}{I_{200}}$$

where I_{200} represents the maximum diffraction intensity of the lattice plane at 2θ = 22–23°, and I_{am} is the minimum diffraction intensity at 2θ = 18–19° (Segal et al. 1959).

K–T solvatochromic parameters measurement of different DESs

The dyes including Nile red (NR), N, n-diethyl-4-nitroaniline (NEt_2), 4-nitroaniline (NH_2), and 4-nitroanisole (OMe) were used to measure K-T solvatochromic parameters. Among them, NR, NEt_2, and NH_2 dye solutions were dissolved in methanol, and the OMe dye solution was dissolved in acetone. After drying, the four dye solutions were mixed with DESs, and then an ultraviolet–visible spectrophotometer (UV-1800, Shimadzu Corporation, Japan) was used for an ultraviolet test with a scanning wavelength of 300–700 nm. The α, β, and π values of different DESs are calculated with the following formula, $\lambda(\#)_{max}$ representing the maximum absorption wavelengths of the dye solutions (# presents NR, NEt_2, NH_2, and OMe dye solutions) (Zhang et al. 2023a; Jessop et al. 2012).

$$\alpha = \frac{19.9657 - 1.0241\pi^* - \nu_{NR}}{1.6078}$$

$$\pi^* = 0.314 \times \left(27.52 - \nu_{NEt_2}\right)$$

$$\beta = 11.134 - \frac{3580}{\lambda\left(NH_2\right)_{\max}} - 1.125\pi^* \quad \text{or}$$

$$\beta = 11.134 - \frac{3580}{\lambda(OMe)_{\max}} - 1.125\pi^*$$

$$\nu_{NR} = \frac{10000}{\lambda(NR)_{\max}} \quad \nu_{NEt_2} = \frac{10000}{\lambda\left(NEt_2\right)_{\max}}$$

Results and discussion

The effect of HBD in DESs on nanofibrillation of cellulose

Understanding the effect of the viscosity of DESs on cellulose behavior is helpful to better use DESs as solvents to treat and transform cellulose materials. By measuring the viscosity of DESs, the interaction between the molecules in DESs and their flow properties can be understood, as well as the physical properties of DESs. What's more, the viscosity of DESs plays an important role in the process of penetration and swelling of cellulose. DESs with high viscosity will be subjected to greater resistance in penetrating cellulose, so the penetration rate is slow. On the contrary, DESs with low viscosity penetrate faster (Yan et al. 2021). DESs composed of LA and other HBAs have low viscosity, which is conducive to the transport and penetration of ions in solvents.

In our previous work, polyethylene glycol (PEG) was used to moisten cellulose to break the hydrogen bond network of cellulose, and then through powerful mechanical processing to prepare cellulose microfibers (CMFs) and cellulose nanofibers (CNFs) (Wang et al. 2017). CNFs, or lignocellulose nanofibers (LCNFs), can also be prepared by the combination of glycerol and a slight acid (0.64% w/w sulfuric acid based on the whole reaction system) system swelling and mechanical treatment (Zhang et al. 2022b). In addition, carboxyl-DESs were excellent solvent systems for the preparation of CNFs (Liu et al. 2021; Wang et al. 2023c). In the preparation process of CNFs, cellulose fibers after solvent pretreatment were more easily stripped off to obtain CNFs than untreated cellulose fibers. The hydrogen bond network of cellulose is destroyed after solvent pretreatment, which makes the preparation of CNFs easy. Solvents seem to play an important role in breaking down hydrogen bond networks of cellulose. Therefore, the key factors for the preparation of CNFs in the DES system were investigated in detail.

Firstly, the effect of LA and ChCl pretreatment on the morphology of prepared cellulose fibers was investigated (shown in Fig. 1a, b). As can be seen from the figure, CMFs can be prepared after LA solution pretreatment combined with mechanical treatment, but the diameter of most cellulose fibers was difficult to reach the nanometer level (Fig. 1a). Analogously, the cellulose fibers pretreated with ChCl

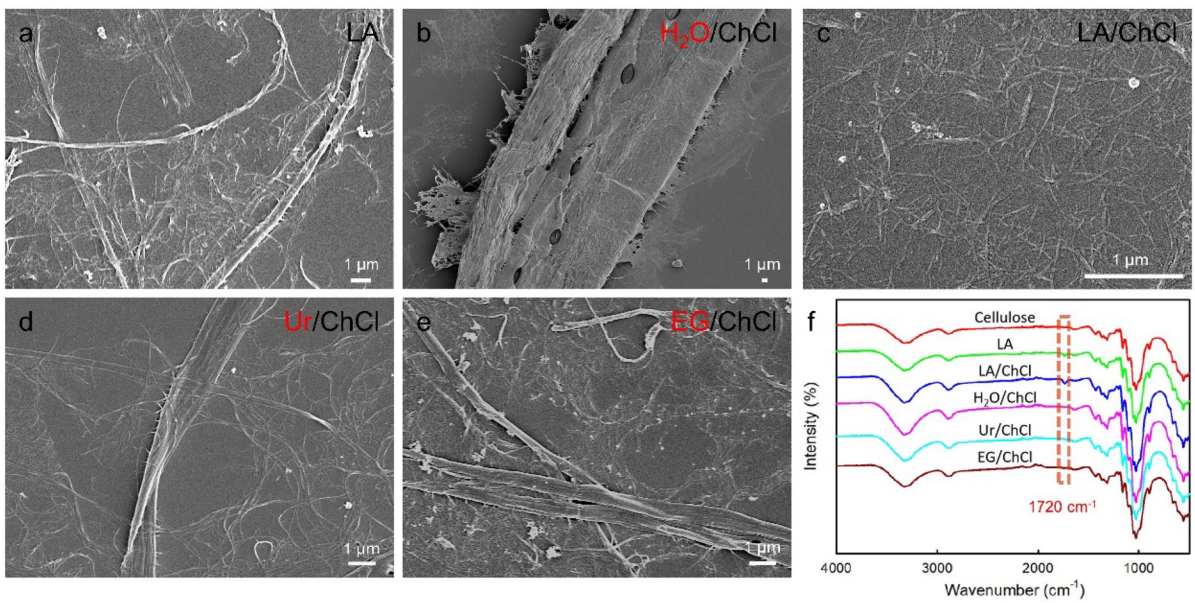


Fig. 1 The micromorphology of cellulose fibers prepared by different solvent systems. (**a**) LA, (**b**) ChCl solution, (**c**) LA/ChCl. DESs composed of different HBD and ChCl, (**d**) Ur/ChCl, and (**e**) EG/ChCl. (**f**) FT-IR spectra of cellulose prepared by different solvent systems

solution maintained the morphology of the cellulose fibers after beating treatment, and the diameter of the cellulose fibers became slightly thin but without significant change (Fig. 1b). It can be seen that the use of LA or ChCl alone is not effective on cellulose nanofibrillation, because the solvent is difficult to resist the dense hydrogen bond network of cellulose and then enter the cellulose interior to promote cellulose nanofibrillation. However, CNFs can be successfully prepared by using LA/ChCl DES pretreatment with mechanical grinding (Fig. 1c).

A high-temperature pretreatment using LA/ChCl DES is beneficial to cellulose nanofibrillation. In the process of swelling, intermolecular hydrogen bonds at a high temperature are dynamic and irregular, and it is easy to form new hydrogen bond sites with LA and Cl^{-1}, weakening the hydrogen bond network of cellulose. Therefore, the cellulose fibril can be peeled by combining the subsequent mechanical treatment to obtain CNFs (Wohlert et al. 2021). After LA/ChCl DES high-temperature swelling and colloid mill grinding, the fiber diameter was reduced to tens of nanometers. It has been reported that the presence of LA/ChCl DES led to esterification modification and effective swelling of cellulose, and the interaction between LA/ChCl DES and cellulose destroyed a large number of hydrogen bonds between molecular chains (Yang et al. 2019; Hong et al. 2020). LA/ChCl DES molecules can efficiently diffuse into the dense structure of cellulose by hydrogen bonding and alter its internal interactions. Among them, Cl^- competed with hydroxyl groups in cellulose to form hydrogen bonds, resulting in a weakening of strong hydrogen bond interactions in cellulose. Therefore, LA/ChCl DES can depolymerize cellulose by breaking hydrogen bonds in molecular chains under mild conditions (Wu et al. 2023). Although few studies have reported that DESs composed of alcohols, urea, amino acids, and ChCl were used for the preparation of CNFs, the effect was generally not up to expectations, and it was only applied to the pretreatment of biomass (Tong et al. 2023; Suopajärvi et al. 2017; Smirnov et al. 2020). The results of this work were roughly the same. As shown in Fig. 1d and 1e, CNFs could not be prepared by treating cellulose in Ur/ChCl and EG/ChCl DES systems, and the fiber diameter remained at the micron level. Only, CMFs were obtained, which also highlighted the advantages of LA/ChCl DES in the preparation of nanocellulose.

The esterification modification of cellulose caused by DESs is also shown in Fig. 1f. It was found that compared to untreated cellulose, the samples treated with LA and LA/ChCl DES showed distinct banding at about 1720 cm^{-1}, which was attributed to the characteristic peak of the carbonyl group. It was proved that the hydroxyl group of cellulose could be esterified with the carboxyl group of LA during LA/ChCl DES pretreatment (Zeng et al. 2023; Lim et al. 2021). Moreover, the peak strength of the cellulose samples obtained by LA/ChCl DES treatment was higher than that of the LA system, which proved that the presence of Cl^- weakened the strong hydrogen bond network of cellulose, making it easier for LA to contact the hydroxyl group of cellulose for esterification modification. After the treatment of cellulose by Ur/ChCl and EG/ChCl DES, the surface groups of cellulose did not change significantly, and no other significant differences were observed in the FT-IR spectrum, which proved that the chemical modification of cellulose was minimal (Xu et al. 2023; Sánchez-Badillo et al. 2022). Therefore, LA can provide more active protons than other systems in the preparation of CNFs, thereby improving the proton mobility of LA/ChCl DES, and breaking the hydrogen bond network of cellulose.

The effect of HBA in DESs on nanofibrillation of cellulose

Hydrogen bonding was one of the most important intermolecularly directed interactions in cellulose. The free hydroxyl group can form a large number of hydrogen bonds, which makes its bond energy become huge. In DESs, HBA and HBD can also form a large number of hydrogen bonds, but their bond energy is low. When DES moistens cellulose and reacts with it, HBD and HBA separate and form a new hydrogen bond network with cellulose, weakening the hydrogen bond network of cellulose itself. Among them, the essence of breaking the cellulose hydrogen bond was to provide a large enough energy to the hydrogen bond, such as heating, mechanical handling, or providing more electronegative HBA molecules (Chen et al. 2023a).

In this work, the effects of anions and cations in HBA on cellulose nanofibrillation were discussed in detail. The effects of different anions on cellulose nanofibrillation are shown in Fig. 2a1–a3. Cellulose

was treated with DES composed of three different anions (bromine ion (Br^-), lactate ions, and dihydrogen citrate ions), and CNFs with an average diameter of less than 100 nm were prepared successfully after treatment using LA/ChBr DES (Fig. 2a1). It is found that Br^- and Cl^- are both halogens, both of them play an important role in opening cellulose chains during cellulose dissolution, and larger Br^- are more conducive to the opening of cellulose chains (Lara-Serrano et al. 2020), thus making the structure of cellulose become soft during pretreatment, and the nanofibrillation of cellulose can be successfully achieved with mechanical treatment. In summary, both Br^- and Cl^- can form new hydrogen bonds with cellulose hydroxyl groups, thus weakening the original hydrogen bonds in the cellulose structure, and making the deconstruction of cellulose fibers simple. However, the LA/Ch-Ca (Fig. 2a2) and LA/Ch-La (Fig. 2a3) DES system did not contain any halogen ions, high temperature swelling treatment and colloid grinding were difficult to break the hydrogen bond network of cellulose, the diameter of the cellulose fibers couldn't be reduced to the nanometer scale after colloid grinding, and there was no significant change compared with cellulose fibril. The reason may be that HBD formed a strong interaction with LA and could not effectively act on the hydrogen bond of cellulose, which strongly limits the role of the DES towards cellulose crystallite (Sánchez-Badillo et al. 2022). In contrast, CNF can be successfully prepared using DESs composed of different cationic systems containing Cl^-.

As shown in Fig. 2b1–b3, the diameter of the obtained CNFs was less than 100 nm, and the length was tens to hundreds of nanometers with deep colloid mill grinding. CNFs can be successfully prepared using LA/NH_4Cl and LA/DTAC DESs. It was worth noting that the diameter of the fibers was very uniform and the shape of the fiber seemed to be very similar to CNC after LA/$ZnCl_2$ DES treatment. Similarly, recycled molten ferric chloride with strong acidity has been reported as an excellent solvent system for the preparation of CNC from hydrolyzed cellulose. It can penetrate the cellulose molecular chain, destroy the hydrogen bond between cellulose molecules, and promote the disintegration of CNCs (Yang et al. 2023). However, molten ferric chloride can cause excessive damage to the cellulose crystal structure and corrode equipment, if the conditions of LA/$ZnCl_2$ DES treatment of cellulose are properly regulated, this may be a very good system for preparing CNCs. Therefore, the key role of anions in cellulose nanofibrillation was confirmed, and the presence of Cl^- would disrupt the cellulose hydrogen bond network and contribute to cellulose nanofibrillation. Figure 2c1–c4 shows the TEM images of the CNFs prepared using DESs composed of different cations of HBAs and LA. It can be seen that the diameters of most fibers were between 20 and 100 nm, with lengths of several microns. Moreover, the diameter distribution of CNFs was uniform, and there was no significant difference in CNFs prepared by different DES systems. The results of TEM images were similar to those of SEM images, which proved that these DES were excellent solvents for preparing CNFs. Dynamic light scattering (DLS) analysis is often used to measure the change of light scattering intensity caused by the Brownian motion of the particles in the sample and infer the size of the particles from it, and it is often used to measure the particle size of nanocellulose (Sai Prasanna and Mitra 2020; Souza et al. 2020). The Z-avg, mean particle size, and polydispersity index (PDI) obtained by DLS analysis of CNFs prepared by DESs composed of different anions of HBAs and LA are shown in Table 2. Here for CNFs prepared by DESs composed of different anions of HBAs and LA, the Z-avg by DLS analysis obtained to be around 1100 nm with mean particle size calculated to be around 200 nm, and PDI of less than 0.5 indicating uniformity of particle size, and almost no agglomeration in the suspension. Since acid hydrolysis of cellulose during CNF production disrupted glycosidic bonds of cellulose in a very random way resulting in their non-uniform size distribution. It also further decreased the dimensions resulting in uniform particle size distribution with the application of mechanical treatment under DESs for 40 min. Thus, CNFs

Table 2 The Z-avg, mean particle size, and PDI obtained by DLS analysis of CNFs prepared by DESs composed of different anions of HBAs and LA

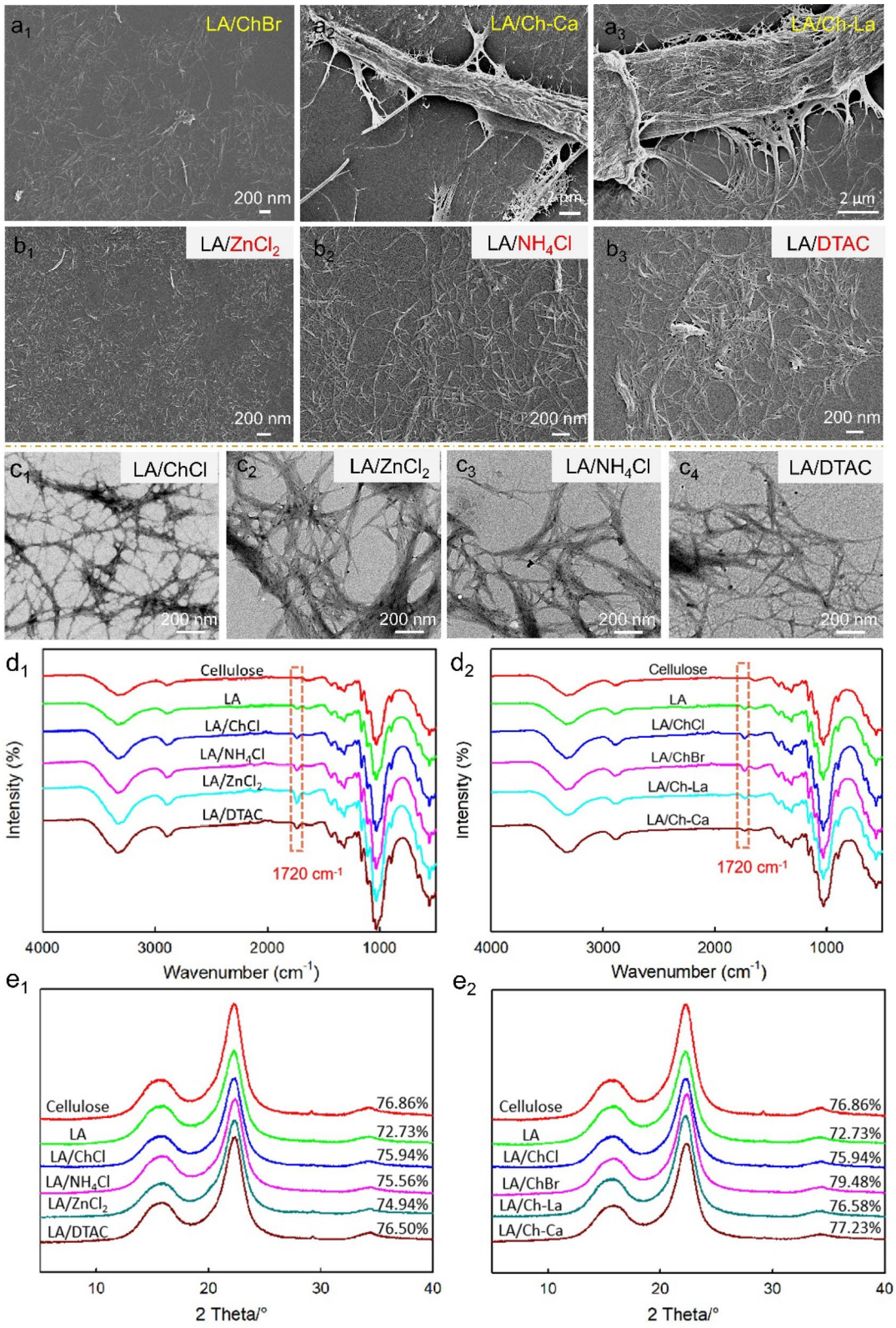
DESs	Z-avg (nm)	mean particle size(nm)	PDI
LA/ChCl	1196.60	312.57 ± 192.31	0.47
LA/NH4Cl	1208.50	294.12 ± 171.33	0.33
LA/ZnCl2	1098.89	273.84 ± 162.28	0.29
LA/DTAC	1220.81	288.09 ± 187.55	0.46

with uniform size can be successfully prepared by DES pretreatment and further mechanical treatment.

The nanofibrillation of cellulose in carboxylic DES systems is usually accompanied by surface esterification, which also prevents the cellulose from being over-hydrolyzed to obtain high-yield CNFs (Liu et al. 2021). The FT-IR spectra of cellulose samples also showed the esterification modification of cellulose by DESs (Fig. 2d). The main peak of O−H tensile vibration is 3428 cm^{-1}. At 2901 cm^{-1}, 2856 cm^{-1}, and 897 cm^{-1}, a series of vibration peaks appear corresponding to the deformation vibration of C−H in cellulose. Peaks at 1429 cm^{-1}, 1161 cm^{-1}, 1110 cm^{-1}, 1059 cm^{-1}, and 1033 cm^{-1} are typical for cellulose (Zhang et al. 2022a). In addition, a distinct band appeared at 1720 cm^{-1}, which was the characteristic peak of the carbonyl group. It has been proved that zinc ions, aluminum ions, and other transition metal elements can catalyze the esterification reaction (Hong et al. 2019; Yuan et al. 2023). It could be seen from the figure, that the DES systems containing LA had esterification reactions in the high-temperature swelling treatment of cellulose. In addition, the peak strength of ester bond peaks in DESs composed of different cations (LA/ChCl, LA/$ZnCl_2$, LA/NH_4Cl, and LA/DTAC.) was higher than that in pure LA systems (Fig. 2d1), and the ester bond peak strength of cellulose treated with LA/$ZnCl_2$ DES system was the highest due to the catalytic action of metal zinc ions. Similarly, an obvious characteristic peak appeared at 1720 cm^{-1} due to the key role of Br^- in the LA/ChBr (Fig. 2d2), indicating that the system was also highly accessible to cellulose. Therefore, the key role of anions was confirmed, more hydroxyl groups were exposed after the cellulose chain was opened, and the esterification reaction with LA was more accessible, which led to a significant increase in the carbonyl peak in the FT-IR spectrum. However, the ester bond peak strength after the treatment of LA/Ch−Ca and LA/Ch−La DES systems was very weak, which also reflected the small effect of the system on cellulose, ultimately limiting the cellulose nanofibrillation. It was concluded that anions played an important role in destroying the hydrogen bond network of cellulose and increasing the accessibility of solvents to cellulose.

In addition, the amorphous regions of cellulose were easily removed by DESs during pretreatment, which might increase the overall crystallinity of cellulose (Kumar et al. 2015; Loow et al. 2018). In the latest study, some DESs were used to dissolve cellulose, in which cellulose underwent a type I to type II transformation (Zhong et al. 2022). In our work, the cellulose raw material comes from the bleached hardwood pulp board produced by the factory. The alkali treatment at a high temperature (165℃) and bleaching treatment in the pulping process removed most of the amorphous areas in the cellulose (Hafid et al. 2021). Moreover, it is reported that only internal or external fibrillation occurred with DES treatment without a great extent of modification in terms of chemical composition and crystallinity (Mnasri et al. 2022). In general, the pattern that exhibited diffraction peaks at 2θ = 15.5° (110), 22.7° (200) and 34.7° (004) corresponded to a natural type I crystal structure. The emergence of new peaks at 2θ = 11.8° and 21.1° in the regenerated cellulose pattern demonstrated the reconstruction of hydrogen bonds after the regeneration process (Tong et al. 2022). In this work, the aggregation structure of the CNFs obtained by all DES systems remains unchanged, as demonstrated by the XRD curves (shown in Fig. 2e1–e2). The pattern exhibited diffraction peaks at 2θ = 15.5° (110), 22.7° (200), and 34.7° (004), indicating the CNFs obtained by all DES systems maintained the natural cellulose type I structure, and the crystallinity of cellulose did not change significantly (about 75%), indicating that the crystal structure of cellulose was not broken by different DES treatment. It was also proved that different anions and cations in this study had similar effects on the crystal structure of cellulose, and the obtained CNFs still had high crystallinity.

XPS technology can provide qualitative and quantitative analysis of the elements and chemical states on the surface of the sample and has a strong ability to analyze the surface groups of cellulose materials (Greczynski et al. 2023). The high-resolution XPS scans over C (1 s) peaks of the cellulose samples treated by different solvent systems are shown in Fig. 3. Theoretically, pure cellulose exhibits only two carbon signals in C (1 s) XPS spectra: C−O (alcohols and ethers) at a binding energy of 286.0 eV and O−C−O (acetal) moieties at around 287.8 eV (Fig. 3a1). The binding energies with peaks at 284.8, 286.6, 287.8, and 288.2 eV were attributed to C−C/C=C groups, C−O/C−OH groups, O−C−O groups, and O−C=O groups, respectively (Zhang et al. 2020; Wang et al. 2023a).


a1
LA/ChBr
200 nm
a2
LA/Ch-Ca
2 μm
a3
LA/Ch-La
2 μm
b1
LA/ZnCl2
200 nm
b2
LA/NH4Cl
200 nm
b3
LA/DTAC
200 nm
c1
LA/ChCl
200 nm
c2
LA/ZnCl2
200 nm
c3
LA/NH4Cl
200 nm
c4
LA/DTAC
200 nm
d1
Cellulose
LA
LA/ChCl
LA/NH4Cl
LA/ZnCl2
LA/DTAC
1720 cm-1
Intensity (%)
4000
3000
2000
1000
Wavenumber (cm-1)
d2
Cellulose
LA
LA/ChCl
LA/ChBr
LA/Ch-La
LA/Ch-Ca
1720 cm-1
Intensity (%)
4000
3000
2000
1000
Wavenumber (cm-1)
e1
Cellulose
LA
LA/ChCl
LA/NH4Cl
LA/ZnCl2
LA/DTAC
76.86%
72.73%
75.94%
75.56%
74.94%
76.50%
10
20
30
40
2 Theta/°
e2
Cellulose
LA
LA/ChCl
LA/ChBr
LA/Ch-La
LA/Ch-Ca
76.86%
72.73%
75.94%
79.48%
76.58%
77.23%
10
20
30
40
2 Theta/°

◂**Fig. 2** The micromorphology of cellulose fibers prepared by different solvent systems. DESs composed of different anions of HBAs and LA, (**a**$_1$) LA/ChBr, (**a**$_2$) LA/Ch−Ca, (**a**$_3$) LA/Ch−La, and DESs composed of different cations of HBAs and LA, (**b**$_1$) LA/$ZnCl_2$, (**b**$_2$) LA/NH_4Cl and (**b**$_3$) LA/DTAC. (**c**) TEM images of CNFs prepared by DESs composed of different cations of HBAs and LA, (**c**$_1$) LA/ChCl, (**c**$_2$) LA/$ZnCl_2$, (**c**$_3$) LA/NH_4Cl and (**c**$_4$) LA/DTAC. (**d**) FT−IR spectra of cellulose fibers prepared by different cation (**d**$_1$) and anion (**d**$_2$) DES systems. (**e**) XRD spectra of cellulose prepared by different cation (**e**$_1$) and anion (**e**$_2$) DES systems

The binding energy of the cellulose treated with EG/ChCl and Ur/ChCl systems did not change significantly, indicating that the groups on the cellulose surface did not change (Fig. 3a2-a3), and the same phenomenon also appeared in the H_2O/ChCl system (Fig. 3b2

). However, The O−C=O groups can be detected by esterification on the surface of cellulose with LA at a high temperature (Fig. 3b1). The same reaction also occurred in LA/ChCl, LA/ChBr, LA/$ZnCl_2$, LA/NH_4Cl, and LA/DTAC DES systems (Fig. 3c1, d1-d3), indicating that chloride ions could destroy the hydrogen bond network inside and outside the cellulose molecules during the pretreatment, thus more hydroxyl groups were exposed on the cellulose surface for esterification, which protected the cellulose from excessive hydrolysis and promoted the further nanofibrillation of cellulose. The degree of substitution (DS) of CNFs prepared by LA/ChCl, LA/ChBr, LA/$ZnCl_2$, LA/NH_4Cl, and LA/DTAC DES systems were 0.55, 0.56, 0.73, 0.52, 0.66, respectively. The DS of cellulose obtained by LA treatment alone was only 0.25 (seen in Table 3). It has also been shown that anions have an important contribution to the destruction of cellulose hydrogen bond networks, thereby improving the accessibility of cellulose.

On the contrary, in the LA/Ch-Ca and LA/Ch-La DES systems, although the system contained LA, no O−C=O groups were detected in the cellulose sample (consistent with the results of only weak carbonyl peaks in the two systems in the FT-IR spectrum), indicating that the hydrogen bond network of cellulose couldn't be destroyed in this system (Fig. 3c2–c3). As a result, LA molecules cannot penetrate the interior of cellulose for esterification, and can only undergo a very limited esterification reaction with the hydroxyl group on the surface of cellulose, and ultimately, the cellulose cannot be nanodispersed. It can be seen that Cl^- played an important role in the process of cellulose nanofibrillation, weakening the hydrogen bond network of cellulose, and LA could modify the hydroxyl group on the surface of cellulose to increase the dispersion stability of CNFs, and at the same time realized the nanofibrillation and surface modification of cellulose under DES systems.

The effect of hydrogen bond alkalinity of DES on cellulose nanofibrillation

There seem to be many indications that the synergistic effects of chemical reactions and non-covalent interactions favor DESs in regulating CNF formation. Some researchers had analyzed the possible mechanism of the formation of nanocellulose regulated by DESs by using molecular dynamics simulation and quantum chemistry calculation techniques and speculated that the hydrogen bond and van der Waals force interaction between DES and cellulose were the key factors of cellulose nanofibrillation (Bi et al. 2023; Wang et al. 2023b). Kamlet-Abboud-Taft parameter (KAT value) was often used to calculate the solvent polarity, including hydrogen bond acidity (α), hydrogen bond alkalinity (β), and dipolarity/polarizability (π^*) (Eyckens et al. 2016; Zhang et al. 2023b). In this case, the β value obtained in the solvatochromic study might be a good indicator of the ability to dissolve cellulose, since high hydrogen bond alkalinity weakened intermolecular and intramolecular hydrogen bonds in cellulose crystals (Abe et al. 2010). The theory is also applied to DESs of cellulose dissolution. It was very promising to select components with high hydrogen bond alkalinity anions, such as Cl^-, OAc^-, $HCOO^-$, etc., to form DESs for cellulose dissolution. Cellulose nanofibrillation disrupts the aggregation structure of cellulose through the destruction of its hydrogen bond network, which leads to a more uniform dispersion of cellulose particles in the solvent. Nevertheless, the dissolution of cellulose involves breaking the hydrogen bond network of cellulose to change the interactions between cellulose molecules, enabling cellulose to interact with solvents and form a more stable solution. Similar to cellulose dissolution, weakening intermolecular and intramolecular hydrogen bonds in cellulose fibers also appears to be the first step in cellulose nanofibrillation. Therefore,

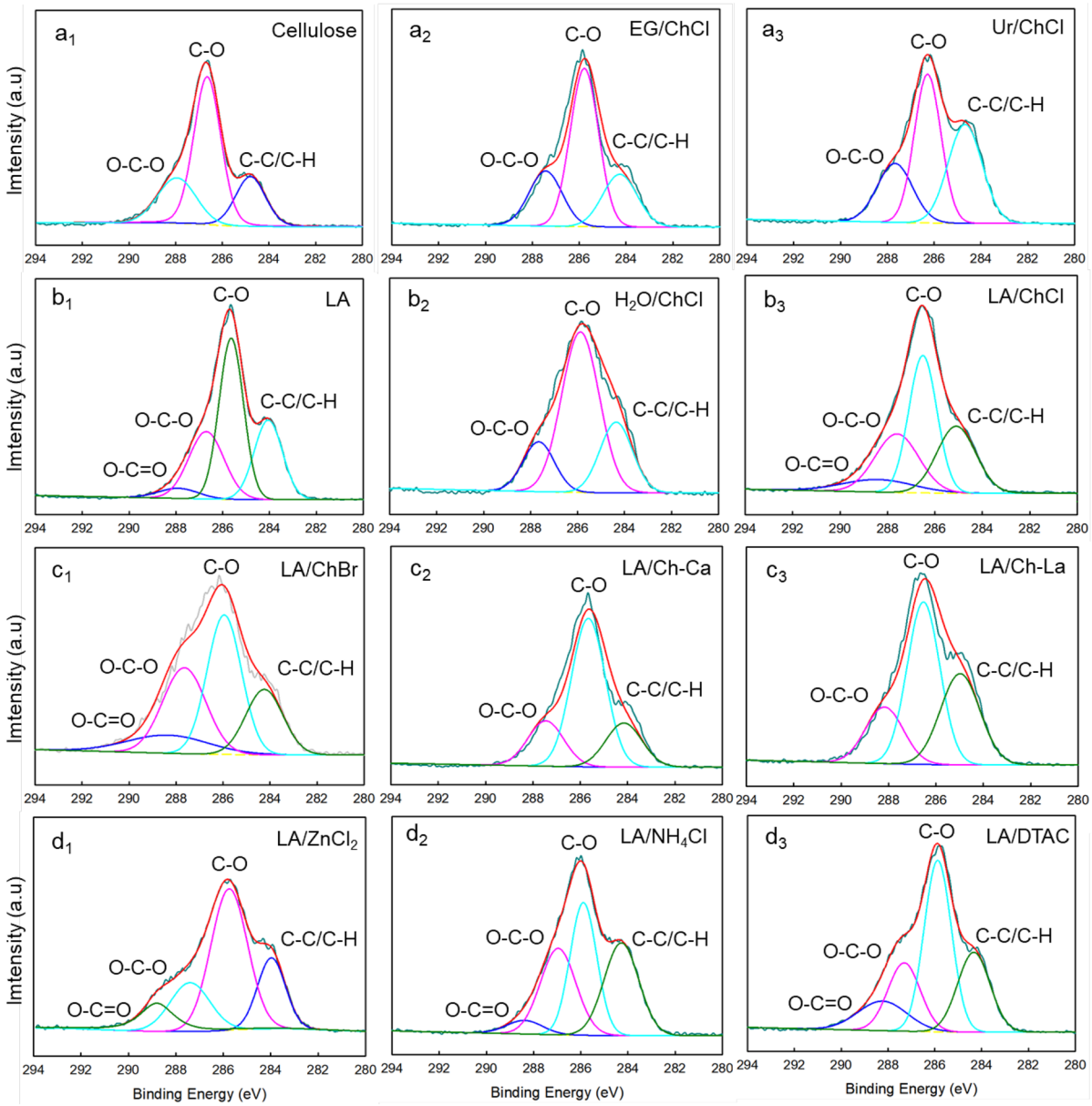


Fig. 3 High-resolution XPS scans over C (1 s) peaks of the cellulose samples treated by different solvent systems. (**a**$_1$) cellulose, (**a**$_2$) EG/ChCl, (**a**$_3$) Ur/ChCl, (**b**$_1$) LA, (**b**$_2$) H_2O/ChCl, (**b**$_3$) LA/ChCl, (**c**$_1$) LA/ChBr, (**c**$_2$) LA/Ch−Ca, (**c**$_3$) LA/Ch−La, (**d**$_1$) LA/$ZnCl_2$, (**d**$_2$) LA/NH_4Cl, and (**d**$_3$) LA/DTAC

DESs with high hydrogen bond alkalinity are very promising for cellulose nanofibrillation.

The Kamlet-Taft empirical parameter curves for different DES systems in different dyes are shown in Fig. 4. The absorption peaks of different DES systems in NR (Fig. 4a), NEt_2 (Fig. 4b), Ome (Fig. 4c1) and NH_2 (Fig. 4c2) were mainly concentrated at 550−670 nm, 400−450 nm and 300−350 nm, respectively (corresponding calculation results are shown in Table 4). It could be calculated that DESs formed by LA and halogen elements had high β values (> 0.5). Among them, the β value of the system containing Cl^- was slightly higher than that of the system containing Br^-, indicating that the system had a higher

Table 3 Element concentration and relative carbon composition from C (1 s) peak analyses determined by the XPS method in the surface layer of cellulose samples treated by different solvent systems

Samples	Carbon group concentration (at.%)				DS
	C–C (284.0 eV)	C-O (285.6 eV)	O-C-O (287.0 eV)	O-C=O (288.2 eV)	
Cellulose	27.63	47.26	25.11	–	–
EG/ChCl	29.35	48.24	22.41	–	–
Ur/ChCl	35.86	42.08	22.10	–	–
LA	25.19	43.79	26.60	4.42	0.25
H_2O/ChCl	15.87	72.22	11.91	–	–
LA/ChCl	25.13	40.01	25.90	8.96	0.55
LA/$ZnCl_2$	28.42	39.50	20.54	11.55	0.73
LA/NH_4Cl	30.17	37.28	24.83	7.72	0.52
LA/DTAC	23.68	42.74	22.33	11.25	0.66
LA/ChBr	21.60	39.57	29.98	8.85	0.56
LA/Ch-La	22.43	55.15	22.42	–	–
LA/Ch-Ca	19.29	61.02	19.69	–	–

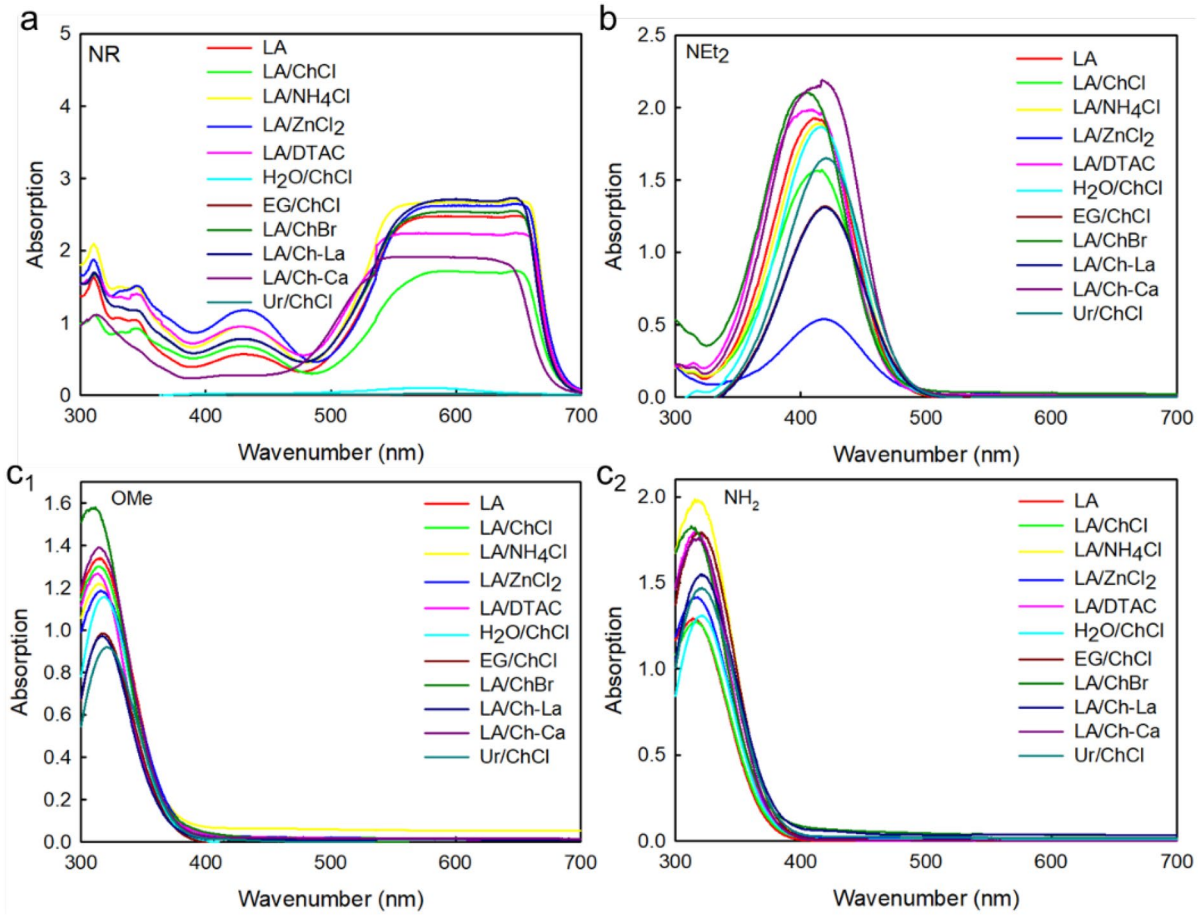


Fig. 4 The Kamlet-Taft empirical parameter curves for different DES systems in different dyes. (**a**) NR, (**b**) Net_2, ($\mathbf{c_1}$) OMe and ($\mathbf{c_2}$) NH_2

Table 4 The hydrogen bond acidity (α), hydrogen bond alkalinity (β), and dipolarity/polarizability (π^*) values of different DESs

Samples	α	π*	β
LA	2.155	1.020	–
LA/ChCl	2.135	1.075	0.606
LA/NH_4Cl	2.263	1.057	0.642
LA/$ZnCl_2$	2.141	1.111	0.609
LA/DTAC	2.209	0.983	0.705
Ur/ChCl	–	1.183	0.416
EG/ChCl	–	1.422	0.360
LA/ChBr	2.107	1.129	0.522
LA/Ch-La	1.892	1.166	0.352
LA/Ch-Ca	1.849	1.093	0.403

hydrogen bond acceptance capacity. The DESs synthesized by LA and chloride had a relatively high α value, indicating that a strong hydrogen bond was formed between the components of DESs, and the strong attraction of Cl^- to the hydrogen proton in LA led to the delocalization of the hydrogen proton, increasing the hydrogen bond acidity of DESs. It could be concluded that DES composed of LA and chloride had a high β value, which indicated that DES had a high hydrogen bond acceptance capacity to strip the hydroxy hydrogen from cellulose, and LA made the system have a high α value, allowing the glycosidic bond of cellulose to be hydrolyzed and broken, thus reducing the size of cellulose. When DES composed of LA and chloride reacted with cellulose, the original hydrogen bond network of cellulose was greatly changed, and a new hydrogen bond structure was formed. DES molecules easily penetrated cellulose to moisten and esterify the cellulose, and cellulose chains were constantly being opened up, thus making the cellulose nanofibrillation process simple (the deconstructed CNFs can be seen in Fig. 2a1, b1-b3).

It has been reported that DES composed of carboxylic acid and ChCl can successfully prepare CNFs, which is an excellent system for preparing CNFs, and can modify cellulose during the nanofibrillation of cellulose (Liu et al. 2021; Wang et al. 2023c). In this study, we demonstrated that carboxylic acid (in the case of LA) and Cl^- played a key role in the cellulose nanofibrillation process, the hydrogen bond network of cellulose was easily broken by DESs containing LA and Cl^-. DES with Cl^- had a high β value, which made it easy to form competitive hydrogen bonds with the hydroxyl group of cellulose when cellulose was swelled by DESs at a high temperature, and changed the original hydrogen bond network of cellulose. LA molecules will randomly esterify with the hydroxyl group on the surface of cellulose, and the ionized hydrogen protons will randomly break the glucoside bond of cellulose, making the cellulose length shorter and the fibers moistened, fluffy, and soft. After subsequent mechanical treatment, CNFs can be successfully prepared.

Therefore, the cellulose nanofibrillation process in DESs (composed of LA and chloride) can be roughly explained as: (a) The original cellulose contains a large number of hydrogen bonds. (b) When cellulose is swelled by DES at a high temperature, Cl^{-1} can penetrate cellulose fibers and form new hydrogen bonds with the hydroxyl group on the surface of cellulose and LA can esterify the exposed hydroxyl group on the surface of cellulose. (c) Hydrating hydrogen protons can hydrolyze the glucoside bond of cellulose randomly in an appropriate amount (usually the b and c processes are carried out simultaneously). (d) Finally, the cellulose fibers were softened, prompting subsequent mechanical treatment to prepare CNFs with surface esterification (the schematic presentation and possible mechanism diagram of the CNFs preparation process from cellulose treated by DESs are shown in Fig. 5).

Conclusions

The synergistic effect of a chemical reaction and non-covalent interaction was favorable for DESs to regulate the formation of CNFs. In this work, it was proved that anion played an important role in the preparation of nanocellulose in DES systems. The experimental results showed that only when LA and Cl^- were present, CNFs could be successfully prepared, the diameter of the fiber was less than 100 nm, and the natural cellulose type I structure was still maintained. It could be seen from FT-IR and XPS curves that the presence of Cl^- increased the DS of cellulose by LA (from 0.25 to more than 0.5). The

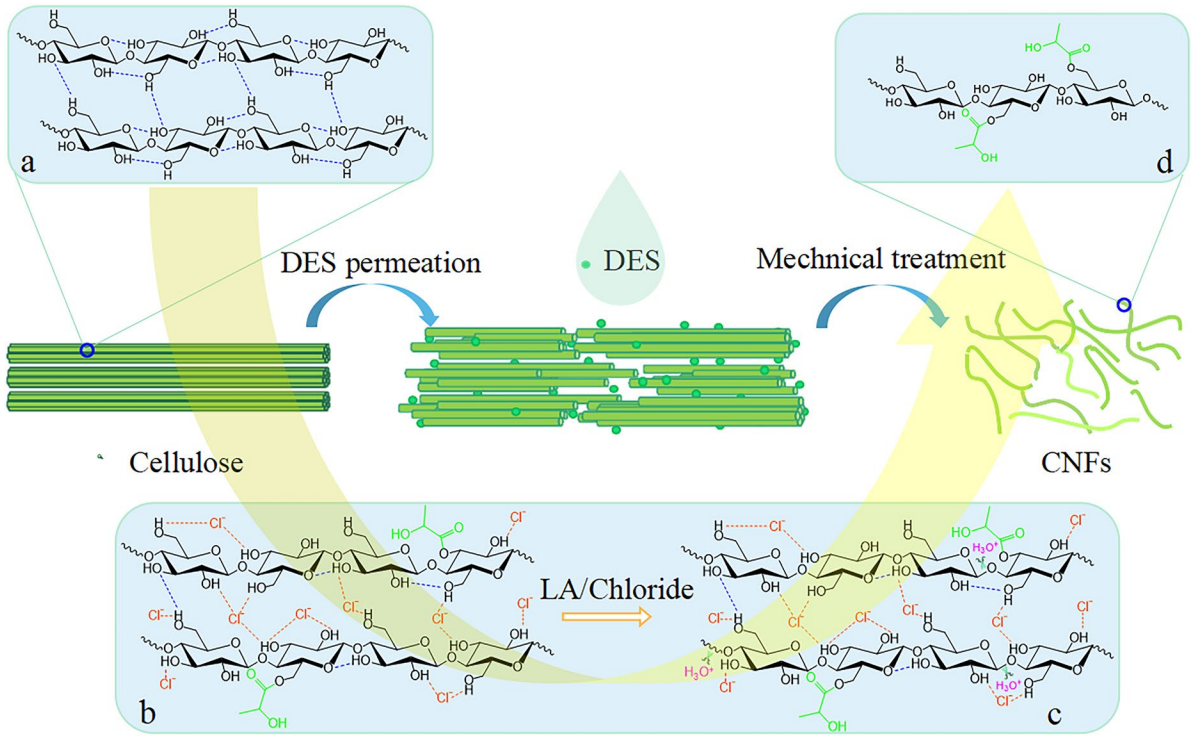


Fig. 5 Schematic presentation and possible mechanism diagram of CNFs preparation process from cellulose treated by DESs

KAT value showed that LA/ChCl, LA/ChBr, LA/$ZnCl_2$, LA/NH_4Cl, and LA/DTAC DES systems had higher β values than others. When Cl^- was present in large quantities inside and outside the cellulose molecules, it was more likely to form new hydrogen bond structures with cellulose and destroy the hydrogen bond network of cellulose, which enabled CNFs to be successfully prepared. This work provided theoretical guidance for selecting solvent systems for preparing nanocellulose and dissolving cellulose.

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Author Contribution Qing Zhang: Investigation, Data Curation, Formal analysis, Writing—original draft. Ziliang Dai: Data Curation, Formal analysis. Lili Zhang: Supervision, Visualization. Zhiguo Wang: Supervision, Project administration, Writing—review & editing.

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Declarations

Conflict of interest The authors declare no competing interests.

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